Shenyu Qin

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Education

Shanghai Jiao Tong University Shanghai, China
John Hopcroft Honors Class, Zhiyuan College Aug 2022 – Jun 2026
B.Eng. in Computer Science and Technology

- GPA: 3.76/4.3 (88.75/100)
- Relevant Coursework: Mathematical Analysis(H), Linear Algebra(H), Combinatorics, Probability, Optimization Methods, Machine Learning, Online Algorithms, Topics in Modern Algorithms, Computational Complexity, Statistical Learning Theory

Research Experience

Shanghai Jiao Tong University Shanghai, China
Advised by Chihao Zhang and Yuhao Zhang Jan 2026 – Present

- Analyzed the approximation guarantee of greedy algorithms for the non-adaptive ProbeMax problem and proved a ratio exceeding $1 - 1/e$.

DIMACS, Rutgers University Piscataway, NJ
Advised by Kangning Wang and Lirong Xia Aug 2025 – Dec 2025

- Studied the existence of α-winning and α-dominating committees of size k in an impartial culture when the number of candidates m is much larger than the number of voters n.
- Established two sharp thresholds for α: an α-winning committee exists w.h.p. iff $\alpha < \frac{k-1}{k}$, and an α-dominating committee exists w.h.p. iff $\alpha < \frac{k-1}{2k}$, when m is sufficiently larger than n and $n = \omega(1)$.
- Wrote a manuscript based on the results.

Shanghai Jiao Tong University Shanghai, China
Advised by Yuhao Zhang Jul 2024 – Mar 2025

- Studied the online load balancing problem for random order inputs, aiming to close the gap between the $\Omega(\sqrt{\log m})$ lower bound and the $O(\log m / \log\log m)$ upper bound by designing a new algorithm, where m is the number of machines.
- Investigated the online graph balancing problem, which is a special case of the original problem, trying to analyze the competitive ratio of a newly-designed algorithm in this case.
- Participated in TCS reading group every week.

Publications

Authors are listed alphabetically unless otherwise noted.

Winning in the Limit: Average-Case Committee Selection with Many Candidates [arXiv]
Yifan Lin, Shenyu Qin, Kangning Wang and Lirong Xia
In *Proceedings of the 27th ACM Conference on Economics and Computation* (EC 2026)

Teaching Assistantships

Algorithm Design and Analysis (CS3950), Shanghai Jiao Tong University Spring 2026

Honors and Awards

Zhiyuan Honors Scholarship (Top 10%), Shanghai Jiao Tong University 2022, 2023, 2024, 2025

Skills

Programming C++, C, Python, Rust

Tools: LaTeX, MATLAB, Coq

Languages: Mandarin (Native), English (TOEFL: 104)